Communication-Free Multi-Shepherd Coordination for Heterogeneous Swarm Deployment in Multi-Region Search and Rescue

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***Abstract*—Disaster reconnaissance operations require robots to inspect separated regions under unreliable communication conditions. This work presents a heterogeneous search-and-rescue architecture in which a small number of capable leaders coordinate numerous inexpensive sensor-equipped followers. A single initialization phase assigns balanced follower subgroups and leaders to designated regions of interest; subsequently, robots do not exchange states, maps, assignments, or commands. Followers integrate local cohesion, separation, leader-avoidance, and obstacle-avoidance behaviors, while leaders operate a collecting, shedding, and driving finite-state controller. Co-shepherds select distinct targets by applying an identical deterministic rule to a shared onboard observation snapshot. The mission, assignments, sampled-data controllers, safety constraints, and observation conditions that ensure consistent target allocation are formalized. This framework deploys compact monitoring subgroups to multiple search regions, enabling local scanning to commence.**

***Index Terms*—shepherding, robotics, swarm, communication-free, SAR**

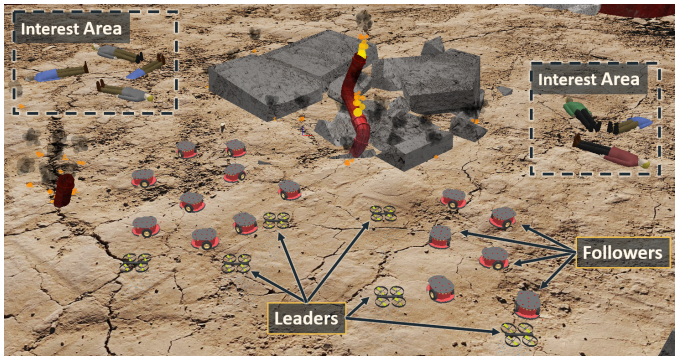


Fig. 1. Conceptual representation of the proposed heterogeneous SAR architecture, in which leaders guide sensor-equipped follower subgroups toward distinct regions of interest under communication-free execution.

I. Introduction

Search-and-rescue (SAR) teams are required to locate survivors and assess hazards in hazardous, partially mapped, and dynamic environments. Field studies have demonstrated the value of robotic sensing [1] as well as the coordination challenges that arise during large-scale deployments [2]. While rescue robots increasingly support reconnaissance, achieving multi-region coverage remains challenging under constraints of time, energy, and communication [3].

Robot swarms enable distributed sensing through scalable local interactions [4]. Techniques such as frontier exploration [5], coordinated mapping [6], and coverage control [7] offer principled mechanisms for deploying mobile sensors. In SAR contexts, swarms have integrated navigation, rendezvous, and victim detection [8], and have conducted parallel searches for multiple targets [9]. Most existing methods treat robots as equivalent decision makers or require the exchange of maps, frontiers, beliefs, or assignments. Heterogeneous teams concentrate strategic computation and perception in a small number of capable robots, while numerous inexpensive robots provide distributed measurements [10]. Local flocking rules generate cohesion and separation without centralized commands [11], and control-theoretic extensions offer formal models of collective motion [12]. However, flocking alone does not address how a limited number of sophisticated robots should divide and transport a large monitoring swarm across separated SAR regions.

Shepherding provides an interaction mechanism by steering a group through its local response. Robotic experiments have demonstrated automatic swarm control [13], and subsequent research has formalized collection and driving behaviors [14]. Alternating between collecting stragglers and driving a compact group enables this approach to scale to larger swarms [15]. Related control formulations address non-cooperative herds [16], multiple target-steering robots [17], and multiple shepherds [18]. Although multiple leaders can accelerate deployment, uncoordinated roles may result in pursuing the same follower, interfering along the same trajectory, or mixing nearby subgroups.

In communication-constrained exploration, robots are typically coordinated through intermittent rendezvous [19] or by maintaining network connectivity during motion [20]. These constraints can delay or restrict deployment in hazardous environments. Most shepherding methods, however, assume a single destination, centralized configuration access, or explicit

coordination among shepherds. This creates a gap and the need for a method that allocates a single swarm across multiple regions while enabling several leaders to coordinate solely through observation.

This work addresses this problem by introducing a heterogeneous SAR architecture. A one-time global initialization partitions sensor-bearing followers and assigns leaders and regions. Subsequently, each robot relies exclusively on onboard relative observations, preloaded mission data, and local control; no states, maps, assignments, or commands are transmitted. Leaders separate the initial swarm, collect dispersed members, and drive compact subgroups to their designated monitoring regions.

The conceptual architecture is shown in Fig. 1. A heterogeneous team of capable leaders and sensor-equipped followers is deployed in a disaster environment containing obstacles and multiple regions of interest. The leaders organize and guide follower subgroups toward their assigned regions, where the followers subsequently perform local monitoring tasks.

The main contributions of this work are as follows: i) a formalization of the multi-region SAR deployment problem incorporating balanced allocation, local sensing, safety constraints, and communication-free execution; ii) a one-time procedure for assigning followers and one or more leaders to each region; iii) a sampled-data controller that integrates local follower motion with leader collection, subgroup shedding, driving, and repulsion; and iv) a deterministic division-of-labor rule and a common-snapshot condition that together yield conflict-free co-shepherd targets without communication.

II. Problem Statement

Let the compact planar workspace be $\mathcal{W} \subset \mathbb{R}^2$, with closed obstacle and hazard sets $\mathcal{O}^{\mathrm{obs}}$ and $\mathcal{O}^{\mathrm{haz}}$. Let, $\mathcal{O} = \mathcal{O}^{\mathrm{obs}} \cup \mathcal{O}^{\mathrm{haz}}$ and, for nonempty $\mathcal{A} \subset \mathbb{R}^2$, $\mathrm{dist}(\mathbf{x}, \mathcal{A}) = \inf_{\mathbf{y}\in\mathcal{A}} \|\mathbf{x} - \mathbf{y}\|_2$. For clearance $d_{\mathrm{obs}} > 0$, the admissible robot-center space is $\mathcal{F} = \{\mathbf{x} \in \mathcal{W} \setminus \mathcal{O} : \mathrm{dist}(\mathbf{x}, \mathcal{O} \cup \partial\mathcal{W}) \geq d_{\mathrm{obs}}\}$. We consider $K \geq 2$ pairwise-disjoint regions of interest in the workspace. Let $\mathcal{K} = \{1, \ldots, K\}$ and define region k, with center $\mathbf{c}_k \in \mathcal{F}$ and radius $r_k > 0$, as $\mathcal{B}_k = \{\mathbf{x} \in \mathcal{F} : \|\mathbf{x} - \mathbf{c}_k\|_2 \leq r_k\}$, $k \in \mathcal{K}$, where $\mathcal{B}_k \cap \mathcal{B}_{k'} = \varnothing$ for $k \neq k'$. Upon delivery to $\mathcal{B}_k$, a subgroup begins local coverage and victim detection. This paper addresses its communication-free delivery rather than the subsequent scanning routine.

The team contains N sensor-bearing followers and M capable leaders, with $K \leq M \leq N$. Let $\mathcal{I}_{\mathrm{F}} = \{1, \ldots, N\}$ and $\mathcal{I}_{\mathrm{L}} = \{1, \ldots, M\}$. Follower $i \in \mathcal{I}_{\mathrm{F}}$ has position $\mathbf{p}_i(t) \in \mathcal{F}$, and leader $\ell \in \mathcal{I}_{\mathrm{L}}$ has position $\mathbf{q}_\ell(t) \in \mathcal{F}$. Both classes follow the planning-level dynamics

$$\dot{\mathbf{p}}_i = \mathbf{u}_i^{\mathrm{F}}, \quad \|\mathbf{u}_i^{\mathrm{F}}\|_2 \leq v_{\mathrm{F}}, \qquad \dot{\mathbf{q}}_\ell = \mathbf{u}_\ell^{\mathrm{L}}, \quad \|\mathbf{u}_\ell^{\mathrm{L}}\|_2 \leq v_{\mathrm{L}}, \tag{1}$$

where $\mathbf{u}_i^{\mathrm{F}}(t), \mathbf{u}_\ell^{\mathrm{L}}(t) \in \mathbb{R}^2$ are the planar velocity control inputs of follower i and leader ℓ, respectively, and $v_{\mathrm{F}}, v_{\mathrm{L}} \in \mathbb{R}_{>0}$ are their maximum speeds. We assume $v_{\mathrm{L}} > v_{\mathrm{F}}$ so that leaders can maneuver around and retain the followers they shepherd.

Each robot measures relative positions within a finite sensing radius. Let $\rho_{\mathrm{FF}}, \rho_{\mathrm{FL}}, \rho_{\mathrm{LF}}, \rho_{\mathrm{LL}} \in \mathbb{R}_{>0}$ denote the follower–follower, follower–leader, leader–follower, and leader–leader sensing radii; the first letter of each superscript identifies the observing class, the second the observed class. Follower i observes $\mathcal{N}_i^{\mathrm{FF}}(t) := \{j \in \mathcal{I}_{\mathrm{F}} \setminus \{i\} \mid \|\mathbf{p}_j(t) - \mathbf{p}_i(t)\|_2 \leq \rho_{\mathrm{FF}}\}$ and $\mathcal{N}_i^{\mathrm{FL}}(t) := \{\ell \in \mathcal{I}_{\mathrm{L}} \mid \|\mathbf{q}_\ell(t) - \mathbf{p}_i(t)\|_2 \leq \rho_{\mathrm{FL}}\}$; leader ℓ's sets $\mathcal{N}_\ell^{\mathrm{LF}}(t)$ and $\mathcal{N}_\ell^{\mathrm{LL}}(t)$ are defined analogously against followers and other leaders, respectively.

These neighbor sets, together with local obstacle and boundary measurements, constitute the information available during execution. At the initialization time $t_0 \in \mathbb{R}_{\geq 0}$, a centralized procedure uses one global configuration snapshot to compute and preload the robot assignments. For every $t > t_0$, no robot transmits states, maps, observations, assignments, or commands. Each policy depends only on onboard observations, its stored assignment, and fixed mission data.

This asymmetry, together with correspondingly larger leader sensing radii ($\rho_{\mathrm{LF}}, \rho_{\mathrm{LL}} \gg \rho_{\mathrm{FF}}, \rho_{\mathrm{FL}}$), reflects the heterogeneous hardware assumption underlying the mission: leaders are equipped with superior sensing and mobility relative to the inexpensive, sensor-limited followers.

To quantify the compactness of a follower subgroup, consider any nonempty set $\mathcal{S} \subseteq \mathcal{I}_{\mathrm{F}}$, whose cardinality is denoted by $|\mathcal{S}|$. Its centroid is $\bar{\mathbf{p}}_{\mathcal{S}}(t) := |\mathcal{S}|^{-1} \sum_{i\in\mathcal{S}} \mathbf{p}_i(t)$, and its radius is $R_{\mathcal{S}}(t) := \max_{i\in\mathcal{S}} \|\mathbf{p}_i(t) - \bar{\mathbf{p}}_{\mathcal{S}}(t)\|_2$, i.e., the largest distance from a subgroup member to the subgroup centroid.

Definition 1 (Congregated subgroup): For desired spacing $\delta > 0$ and slack $\lambda \geq 1$, $\mathcal{S}$ is *congregated* at t if

$$R_{\mathcal{S}}(t) \leq \lambda \sqrt{|\mathcal{S}|}\, \delta. \tag{2}$$

The $\sqrt{|\mathcal{S}|}$ factor captures the radius scaling of an approximately uniform planar cluster with spacing δ.

Let $d_{\mathrm{FF}}, d_{\mathrm{FL}}, d_{\mathrm{LL}} > 0$ denote the required clearances between two followers, a follower and a leader, and two leaders, respectively.

For simplicity, we set $\delta = d_{\mathrm{FF}}$ throughout, so the nominal inter-follower spacing used in the congregation criterion (2) coincides with the minimum required follower-follower clearance.

The control objective can now be stated formally.

Problem 1 (Communication-free multi-region swarm deployment): Using the configuration at t_0, construct pairwise-disjoint follower and leader partitions $\{\mathcal{S}_k\}_{k\in\mathcal{K}}$ and $\{\mathcal{L}_k\}_{k\in\mathcal{K}}$ satisfying

$$\bigcup_{k\in\mathcal{K}} \mathcal{S}_k = \mathcal{I}_{\mathrm{F}}, \quad \bigcup_{k\in\mathcal{K}} \mathcal{L}_k = \mathcal{I}_{\mathrm{L}}, \quad \big||\mathcal{S}_k| - |\mathcal{S}_{k'}|\big| \leq 1, \quad |\mathcal{L}_k| \geq 1. \tag{3}$$

and design post-initialization policies satisfying the information restriction such that every $k \in \mathcal{K}$ has a finite delivery time $T_k > t_0$ for which

$$\mathbf{p}_i(T_k) \in \mathcal{B}_k \quad \forall i \in \mathcal{S}_k, \qquad R_{\mathcal{S}_k}(T_k) \leq \lambda \sqrt{|\mathcal{S}_k|}\, \delta, \tag{4}$$

while all robots remain in $\mathcal{F}$, obey (1), and maintain the three pairwise clearances throughout the mission.

Remark 1 (Implementation criterion): In the reported experiments, mission completion for region k is verified via the practical proxy $\|\bar{\mathbf{p}}_{\mathcal{S}_k}(t) - \mathbf{c}_k\|_2 < \varepsilon$ (subgroup centroid

within tolerance ε of the goal center), rather than checking (4) for each follower directly. This proxy is a necessary but not sufficient approximation of the formal delivery condition; for ε small relative to r_k and under the congregation guarantee of (2), individual follower positions remain close to the centroid, making the proxy a reasonable stand-in.

We assume each region can contain its congregated subgroup, collision-free paths to all regions exist, and the leaders' sensing and speed advantages suffice to retain their followers.

III. Communication-Free Shepherding Methodology

The method has two phases. Once the staged followers form one congregated swarm, their configuration defines t_0 and the fixed subgroup, leader, and region assignments. During execution, followers respond to sensed neighbors and obstacles, while leaders use stored goals and observations to separate, drive, and regroup their assigned subgroups.

The execution phase is implemented as a sampled-data controller. Let $t_h := t_0 + h\Delta t$, where $h \in \mathbb{Z}_{\geq 0}$ is the sample index and $\Delta t \in \mathbb{R}_{>0}$ is the sampling period. To express the local interaction laws compactly, we introduce two auxiliary functions used by both robot classes. First, the following complementary sigmoid functions convert a scalar input into a smooth activation weight. For $z \in \mathbb{R}$, $\kappa \in \mathbb{R}_{>0}$, and $c \in \mathbb{R}$, let $s^+(z;\kappa,c) := (1+e^{-\kappa(z-c)})^{-1}$ and $s^-(z;\kappa,c) := 1 - s^+(z;\kappa,c)$. In the control laws below, z is typically a normalized distance, κ controls the transition steepness, and c is the switching point. Consequently, s^+ strengthens an interaction as z increases, whereas s^- weakens it. Second, the map $\mathrm{dir} : \mathbb{R}^2 \to \mathbb{R}^2$ extracts the direction of a displacement: $\mathrm{dir}(\mathbf{z}) := \mathbf{z}/\|\mathbf{z}\|_2$ for $\mathbf{z} \neq \mathbf{0}$, and $\mathrm{dir}(\mathbf{0}) := \mathbf{0}$. The zero-vector convention prevents an undefined control direction when two relevant positions coincide.

A. Followers dynamics

At each sample, follower $i \in \mathcal{I}_{\mathrm{F}}$ applies four local behaviors: cohesion to maintain swarm integrity, separation to preserve spacing, leader-induced escape for indirect shepherding, and obstacle avoidance to prevent entry into unsafe areas.

Let us define the neighbor barycenter as $\bar{\mathbf{p}}_i^{\mathrm{nbr}} := |\mathcal{N}_i^{\mathrm{FF}}|^{-1} \sum_{j \in \mathcal{N}_i^{\mathrm{FF}}} \mathbf{p}_j$ when $\mathcal{N}_i^{\mathrm{FF}} \neq \varnothing$, and set $\bar{\mathbf{p}}_i^{\mathrm{nbr}} := \mathbf{p}_i$ otherwise. This ensures the cohesion vector is zero for an isolated follower. To independently tune the three distance-dependent interactions, let $\kappa_a, \kappa_r, \kappa_e \in \mathbb{R}_{>0}$ be the cohesion, separation, and escape sigmoid steepnesses, respectively, and let $c_a, c_r, c_e \in \mathbb{R}$ be their switching points.

Cohesion: The cohesion vector points toward the neighbor barycenter, with an activation that increases as the barycenter displacement grows relative to the sensing radius ρ_{FF}:

$$\mathbf{f}_i^{\mathrm{coh}} := \mathrm{dir}\big(\bar{\mathbf{p}}_i^{\mathrm{nbr}} - \mathbf{p}_i\big)\, s^+\left(\frac{\|\bar{\mathbf{p}}_i^{\mathrm{nbr}} - \mathbf{p}_i\|_2}{\rho_{\mathrm{FF}}}; \kappa_a, c_a\right). \quad (5)$$

Separation: Separation sums pairwise repulsion vectors. Because s^- decreases with its input, closer followers contribute more strongly, and the influence decays as their distance increases relative to the desired spacing δ:

$$\mathbf{f}_i^{\mathrm{sep}} := \sum_{j \in \mathcal{N}_i^{\mathrm{FF}}} \mathrm{dir}(\mathbf{p}_i - \mathbf{p}_j) s^-\left(\frac{\|\mathbf{p}_i - \mathbf{p}_j\|_2}{\delta}; \kappa_r, c_r\right). \quad (6)$$

Leader-induced escape: The leaders do not transmit motion commands to the followers. Instead, a follower moves away from every sensed leader, allowing leader motion to collect and drive the swarm through local interaction alone:

$$\mathbf{f}_i^{\mathrm{esc}} := \sum_{\ell \in \mathcal{N}_i^{\mathrm{FL}}} \mathrm{dir}(\mathbf{p}_i - \mathbf{q}_\ell) s^-\left(\frac{\|\mathbf{p}_i - \mathbf{q}_\ell\|_2}{\rho_{\mathrm{FL}}}; \kappa_e, c_e\right). \quad (7)$$

Obstacle avoidance: The preceding behaviors govern robot-robot interactions but do not account for workspace geometry. We therefore represent locally modeled obstacles and hazards by a smooth potential that can be evaluated from onboard observations. Let $\mathcal{I}_{\mathrm{O}}$ be their finite index set. At a query point $\mathbf{x} = (x, y) \in \mathbb{R}^2$, component $o \in \mathcal{I}_{\mathrm{O}}$ induces the scalar potential $\phi_o : \mathbb{R}^2 \to \mathbb{R}_{>0}$ given by

$$\phi_o(\mathbf{x}) := \frac{g_o}{1 + \left|\frac{x-\mu_x^o}{\sigma_x^o}\right|^{2s_x^o} + \left|\frac{y-\mu_y^o}{\sigma_y^o}\right|^{2s_y^o}}, \quad (8)$$

where $(\mu_x^o, \mu_y^o) \in \mathbb{R}^2$ is the component center, $g_o \in \mathbb{R}_{>0}$ is its amplitude, $\sigma_x^o, \sigma_y^o \in \mathbb{R}_{>0}$ are its spatial scales, and $s_x^o, s_y^o \in \mathbb{R}_{\geq 1}$ are its shape exponents. The aggregate potential $\Phi : \mathbb{R}^2 \to \mathbb{R}_{\geq 0}$ is $\Phi(\mathbf{x}) := \sum_{o \in \mathcal{I}_{\mathrm{O}}} \phi_o(\mathbf{x})$. We therefore define the obstacle-avoidance vector inline as $\mathbf{f}^{\mathrm{obs}}(\mathbf{x}) := -\nabla\Phi(\mathbf{x})$.

The four behaviors are combined through gains $K_a, K_r, K_e, K_o^{\mathrm{F}} \in \mathbb{R}_{\geq 0}$ to form the nominal follower velocity, $\mathbf{v}_i^{\mathrm{F,nom}} := K_a\mathbf{f}_i^{\mathrm{coh}} + K_r\mathbf{f}_i^{\mathrm{sep}} + K_e\mathbf{f}_i^{\mathrm{esc}} + K_o^{\mathrm{F}}\mathbf{f}^{\mathrm{obs}}(\mathbf{p}_i)$, where the gains weight cohesion, separation, escape, and obstacle avoidance, respectively. To enforce the hard speed and clearance requirements, let $\mathcal{U}_{i,h}^{\mathrm{F}} \subseteq \mathbb{R}^2$ denote the nonempty, closed, convex set of velocities that satisfy $\|\mathbf{u}\|_2 \leq v_{\mathrm{F}}$. For a closed convex set $\mathcal{U}$, denote Euclidean projection by $\Pi_{\mathcal{U}}(\mathbf{v}) := \arg\min_{\mathbf{u} \in \mathcal{U}} \|\mathbf{u} - \mathbf{v}\|_2^2$. The applied command is then $\mathbf{u}_i^{\mathrm{F},h} := \Pi_{\mathcal{U}_{i,h}^{\mathrm{F}}}(\mathbf{v}_i^{\mathrm{F,nom},h})$.

B. Initialization Phase

Initialization begins at t_0 once the full follower set $\mathcal{I}_{\mathrm{F}}$ satisfies the congregation test in (2). A single global configuration snapshot generates balanced assignments of followers to regions and leaders to subgroups. These assignments are stored onboard and remain fixed throughout execution.

Given $\hat{\mathbf{c}} := \bar{\mathbf{p}}_{\mathcal{I}_{\mathrm{F}}}(t_0)$ and the region count K, we first define, for every $k \in \mathcal{K}$, the *shedding sub-goal*

$$\mathbf{g}_k^{\mathrm{shed}} := \hat{\mathbf{c}} + \eta\big(\cos\theta_k, \sin\theta_k\big), \qquad \theta_k := \frac{2\pi k}{K}, \quad (9)$$

with $\eta \in \mathbb{R}_{>0}$ a fixed separation offset. The sub-goals are evenly spaced around $\hat{\mathbf{c}}$ and independent of the region centers $\{\mathbf{c}_k\}$; index k is paired with $\mathbf{g}_k^{\mathrm{shed}}$ and hence eventually with $\mathbf{c}_k$ by convention, and this pairing is fixed for the mission.

Balanced subgroup partition: Let $r_N := N \bmod K$. Region $k \in \mathcal{K}$ receives follower capacity $n_k := \lceil N/K \rceil$ if $k \leq r_N$ and

$n_k := \lfloor N/K \rfloor$ otherwise. For every $i \in \mathcal{I}_{\mathrm{F}}$ and $k \in \mathcal{K}$, define $d_{ik} := \|\mathbf{p}_i(t_0) - \mathbf{g}_k^{\mathrm{shed}}\|_2$. Let $d_i^{(1)} \le d_i^{(2)}$ be the two smallest values in $\{d_{ik}\}_{k\in\mathcal{K}}$ and define the preference margin inline as $m_i := d_i^{(2)} - d_i^{(1)}$. Followers are processed in decreasing m_i order, ensuring that the most unambiguous preferences are addressed first. Each follower is then assigned to its nearest region with available capacity. Ties are resolved using follower and region indices. The resulting sets $\{\mathcal{S}_k\}_{k\in\mathcal{K}}$ satisfy $|\mathcal{S}_k| = n_k$ and $\big||\mathcal{S}_k| - |\mathcal{S}_{k'}|\big| \le 1$ for all $k, k' \in \mathcal{K}$.

Leader-to-subgroup assignment: Similarly, let $r_M := M \bmod K$ and assign leader capacity $m_k^{\mathrm{L}} := \lceil M/K \rceil$ if $k \le r_M$ and $m_k^{\mathrm{L}} := \lfloor M/K \rfloor$ otherwise. Since $M \ge K$, every capacity is positive. Subgroups are processed in decreasing m_k^{L} order, with region indices used to resolve ties. Subgroup k selects the m_k^{L} nearest unassigned leaders, measured by $\|\mathbf{q}_\ell(t_0) - \bar{\mathbf{p}}_{\mathcal{S}_k}(t_0)\|_2$. Leader indices are used to resolve any remaining ties. This produces leader sets $\{\mathcal{L}_k\}_{k\in\mathcal{K}}$ and the assignment map $a : \mathcal{I}_{\mathrm{L}} \to \mathcal{K}$, where $a(\ell) = k$ if and only if $\ell \in \mathcal{L}_k$. Each leader stores $\mathcal{S}_{a(\ell)}$, its regional goal $\mathbf{c}_{a(\ell)}$, and the shedding sub-goal $\mathbf{g}_{a(\ell)}^{\mathrm{shed}}$ defined above. Initialization constitutes the only stage in which the method utilizes global configuration information.

C. Decentralized Division of Labor

Leaders assigned to the same subgroup coordinate without broadcasting by relying on a rank assigned once, together with the rest of the initialization data. Leader $\ell \in \mathcal{L}_k$ receives $r(\ell) \in \{0, \dots, |\mathcal{L}_k| - 1\}$, obtained by sorting $\mathcal{L}_k$ by the polar angle of $\mathbf{q}_\ell(t_0)$ about $\bar{\mathbf{p}}_{\mathcal{S}_k}(t_0)$ at assignment time. Because $r(\cdot)$ is preloaded and never recomputed, every leader in $\mathcal{L}_k$ derives the same duty partition from $r(\ell)$ alone, without observing or communicating with co-shepherds at runtime.

We define the assigned followers observed by leader ℓ as $\mathcal{W}_\ell^h := \mathcal{N}_\ell^{\mathrm{LF}}(t_h) \cap \mathcal{S}_{a(\ell)}$, its observed centroid as $\hat{\mathbf{p}}_\ell^h := |\mathcal{W}_\ell^h|^{-1} \sum_{i\in\mathcal{W}_\ell^h} \mathbf{p}_i^h$, and the observed subgroup radius as $\widehat{R}_\ell^h := \max_{i\in\mathcal{W}_\ell^h} \|\mathbf{p}_i^h - \hat{\mathbf{p}}_\ell^h\|_2$ for $\mathcal{W}_\ell^h \neq \varnothing$. A missing assigned follower subsequently causes the congregation tests to fail. Consistent communication-free coordination requires the following observation condition.

Assumption 1 (Common snapshot): At each sample, all leaders in $\mathcal{L}_k$ observe the same poses and identities of every robot in $\mathcal{S}_k \cup \mathcal{L}_k$ and are in the same control mode.

This assumption is reasonable when leader sensing covers the compact subgroup and its shepherding arc; otherwise, consistent target and formation selection is not guaranteed.

Collecting mode. Leader ℓ partitions the full angular range about $\hat{\mathbf{p}}_\ell^h$ into $c := |\mathcal{L}_k|$ equal arcs ordered by rank,

$$\Theta_j := \Big[\theta_0 + j\tfrac{2\pi}{c},\ \theta_0 + (j+1)\tfrac{2\pi}{c}\Big), \qquad j = 0, \dots, c-1, \tag{10}$$

where θ_0 is fixed by the current bearing of rank-0's co-shepherd about $\hat{\mathbf{p}}_\ell^h$, and leader ℓ claims sector $\Theta_{r(\ell)}$. A follower $i \in \mathcal{W}_\ell^h$ is a *straggler* relative to ℓ if its bearing about $\hat{\mathbf{p}}_\ell^h$ lies in $\Theta_{r(\ell)}$ and $\|\mathbf{p}_i^h - \hat{\mathbf{p}}_\ell^h\|_2 > \lambda\sqrt{|\mathcal{W}_\ell^h|}\,\delta$; leader ℓ's target $\tau_h(\ell)$ is the farthest such straggler, held for at least $H_{\min} \in \mathbb{Z}_{>0}$ samples once selected. The desired position is

$$\mathbf{w}_\ell^h := \hat{\mathbf{p}}_\ell^h + \rho_\ell\big(\cos\beta_\ell^h, \sin\beta_\ell^h\big), \qquad \rho_\ell := \widehat{R}_\ell^h + d_{\mathrm{FF}}, \tag{11}$$

where β_ℓ^h is the bearing of $\tau_h(\ell)$ about $\hat{\mathbf{p}}_\ell^h$ clamped to $\Theta_{r(\ell)}$, or the midpoint of $\Theta_{r(\ell)}$ when no straggler is present.

Driving mode. The active anchor $\mathbf{a}_\ell^h \in \{\mathbf{g}_{a(\ell)}^{\mathrm{shed}}, \mathbf{c}_{a(\ell)}\}$ switches per the rule of Sec. III-D. All leaders in $\mathcal{L}_k$ identify the same push direction $\mathbf{d}_k^h := \mathrm{dir}\big(\mathbf{p}_\sigma^h - \mathbf{a}_\ell^h\big)$, where $\sigma := \arg\max_{i\in\mathcal{W}_\ell^h} \|\mathbf{p}_i^h - \mathbf{a}_\ell^h\|_2$ is the follower farthest from the active anchor (held stable analogously to $\tau_h(\ell)$). Leaders arrange on a semicircular arc of radius ρ_k^h, scaling with subgroup extent and c, centered opposite $\mathbf{a}_\ell^h$ along $\mathbf{d}_k^h$; leader ℓ occupies the slot at angular position $r(\ell)$ along the arc,

$$\mathbf{w}_\ell^h := \mathbf{o}_k^h + \rho_k^h\big(\cos\gamma_{r(\ell)}, \sin\gamma_{r(\ell)}\big), \tag{12}$$

with $\mathbf{o}_k^h$ the arc center and $\{\gamma_j\}_{j=0}^{c-1}$ the equally spaced slot angles centered on the bearing of $\mathbf{d}_k^h$.

D. Leader Controller: Finite State Machine

The finite-state machine coordinates subgroup separation, transport, and cohesion recovery.

Leader ℓ regards its subgroup as locally congregated when $\mathcal{W}_\ell^h = \mathcal{S}_{a(\ell)}$ and $\widehat{R}_\ell^h \le \lambda\sqrt{|\mathcal{S}_{a(\ell)}|}\,\delta$. The congregation test is false when $\mathcal{W}_\ell^h = \varnothing$.

Formally, the leader controller has two modes, COLLECTING and DRIVING. Upon initialization, every leader begins DRIVING with active anchor $\mathbf{g}_{a(\ell)}^{\mathrm{shed}}$ — informally, "shedding" — since subgroup congregation was already required to trigger initialization. Once $\|\hat{\mathbf{p}}_\ell^h - \mathbf{g}_{a(\ell)}^{\mathrm{shed}}\|_2 \le \varepsilon_{\mathrm{shed}}$, where $\varepsilon_{\mathrm{shed}} \in \mathbb{R}_{>0}$ is the shedding tolerance, the active anchor switches to $\mathbf{c}_{a(\ell)}$ and driving continues toward the final region. During DRIVING, loss of congregation invokes COLLECTING, whereas restored congregation returns the leader to DRIVING. Satisfaction of (4) terminates shepherding and initiates local monitoring. The desired position $\mathbf{w}_\ell^h$ in each mode is given by (11) (Collecting) or (12) (Driving); both place the leader at a bounded offset from the local group that induces the target's escape response toward the active anchor.

E. Inter-Leader Repulsion

Inter-leader repulsion prevents nearby co-shepherds from obstructing one another while tracking their assigned followers. Let $\kappa_{\mathrm{rep}}, c_{\mathrm{rep}} \in \mathbb{R}_{>0}$ denote the steepness and normalized switching point of the repulsion profile, respectively. The repulsion acting on leader ℓ is

$$\mathbf{f}_\ell^{\mathrm{rep},h} := \sum_{\ell' \in \mathcal{N}_\ell^{\mathrm{LL}}(t_h)} s^-\left(\frac{\|\mathbf{q}_\ell^h - \mathbf{q}_{\ell'}^h\|_2}{d_{\mathrm{LL}}}; \kappa_{\mathrm{rep}}, c_{\mathrm{rep}}\right) \mathrm{dir}\Big(\mathbf{q}_\ell^h - \mathbf{q}_{\ell'}^h\Big). \tag{13}$$

Thus, nearby leaders produce stronger repulsion, whereas distant leaders have little influence. Leaders are modeled as aerial units operating above ground-level obstacles and hazards; consequently, their admissible velocities are constrained only by their own speed, the workspace boundary, and clearance from other leaders, with no obstacle-avoidance term. Let $\mathcal{A}_{\ell,h}^{\mathrm{L}}$

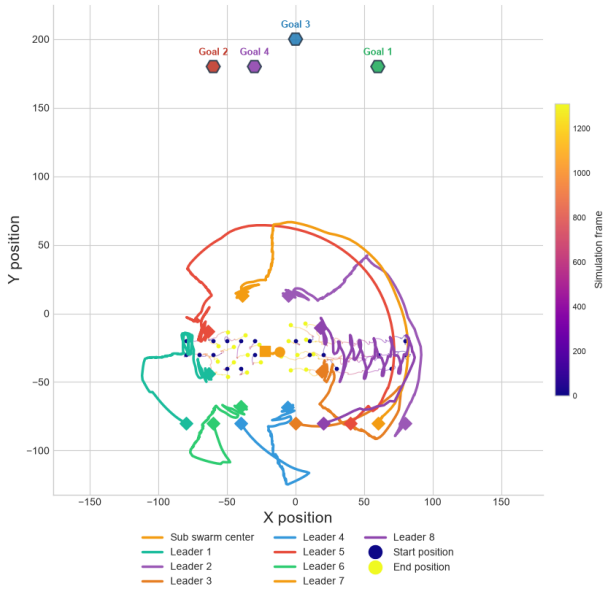


Fig. 2. Congregation phase: leaders move to rank-ordered sector positions and drive stragglers toward the flock center (Sec. III-C). $M = 8$ leaders, $N = 20$ followers, $K = 4$ goals, no obstacles.

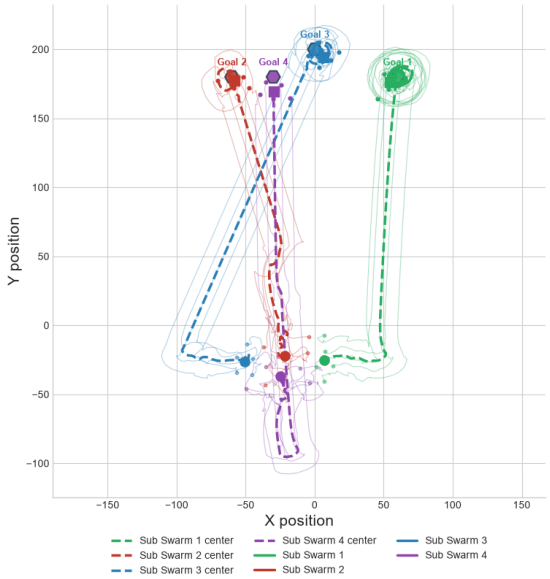


Fig. 3. Driving phase: sub-swarms move toward their shedding sub-goal, then toward their final region (Sec. III-D). $M = 8$ leaders, $N = 20$ followers, $K = 4$ goals, no obstacles.

contain the safe-pointing unit normals of the active robot and boundary clearance constraints. The admissible leader velocities are

$$\mathcal{U}^{\mathrm{L}}_{\ell,h} := \left\{ \mathbf{u} \in \mathbb{R}^2 : \|\mathbf{u}\|_2 \le v_{\mathrm{L}},\ \mathbf{n}^\top \mathbf{u} \ge 0\ \forall \mathbf{n} \in \mathcal{A}^{\mathrm{L}}_{\ell,h} \right\}. \quad (14)$$

The nominal command combines target tracking and inter-leader repulsion:

$$\begin{aligned} \mathbf{v}^{\mathrm{L,nom},h}_{\ell} &:= \mathbf{v}^{\mathrm{trk},h}_{\ell} + K_{\mathrm{rep}} \mathbf{f}^{\mathrm{rep},h}_{\ell} \\ \mathbf{u}^{\mathrm{L},h}_{\ell} &:= \Pi_{\mathcal{U}^{\mathrm{L}}_{\ell,h}} \left(\mathbf{v}^{\mathrm{L,nom},h}_{\ell} \right),\ \mathbf{q}^{h+1}_{\ell} := \mathbf{q}^{h}_{\ell} + \Delta t\, \mathbf{u}^{\mathrm{L},h}_{\ell}, \end{aligned} \quad (15)$$

where $K_{\mathrm{rep}} \in \mathbb{R}_{\ge 0}$ is the repulsion gain, and $\Pi_{\mathcal{U}}$ denotes Euclidean projection onto $\mathcal{U}$. Because $\mathcal{U}^{\mathrm{L}}_{\ell,h}$ is nonempty, closed, and convex, the projection is unique and enforces the speed and clearance constraints.

IV. Case Studies

We present two representative scenarios exercising the constants and control laws introduced above, using the parameter symbols defined in Sec. II and Sec. III-C.

First Case: Division and Shedding Every mission begins with congregation. Each leader takes up a station around the swarm rather than pushing through it, so that no two leaders compete for the same followers the sector partition of (10) keeps their claimed regions disjoint by construction. As shown in Fig. 2, each leader first moves to its holding position and then drives the farthest follower within its own sector toward the swarm center, per the target rule of Sec. III-C. Once the swarm is congregated, shedding begins: the sub-goals $\{\mathbf{g}^{\mathrm{shed}}_k\}_{k \in \mathcal{K}}$ of (9) are placed around the swarm center as can be seen in Fig. 3, pulling each sub-swarm outward toward its assigned region. From this point on, no leader exchanges messages with another coordination relies entirely on onboard sensing of neighbor positions together with the preloaded subgroup membership $\mathcal{S}_{a(\ell)}$, $\mathcal{L}_{a(\ell)}$ (Sec. III-B), which each leader uses to decide whom to collaborate with and which followers to drive.

Second Case: Driving with Obstacle Avoidance Congregation proceeds essentially as in the first case: since no follower starts near an obstacle, the obstacle-avoidance term of the follower force law (Sec. III-C) remains inactive throughout this phase. During driving, followers are guided through the workspace as before; once a follower comes within range of an obstacle, its obstacle-avoidance force term steers it clear, and this holds independently for every affected follower. Leaders adapt to the resulting deviations indirectly: they are modeled as quadcopters operating above the ground-level hazards and therefore never need to avoid the obstacles themselves, but since their desired position is always defined relative to the followers they are driving, a follower's detour is enough to redirect its leader toward the new site, as shown in Fig. 4.

Fig. 5 shows a single instant of the driving formation described by (12): leaders arrange on a semicircular arc opposite the goal, at twice the sub-swarm radius, so that pushing the farthest follower does not require any leader to cross another's path. Because each leader's slot is fixed by its rank $r(\ell)$ rather than recomputed from the live configuration, the formation holds without leaders needing to observe or negotiate with one another at runtime.

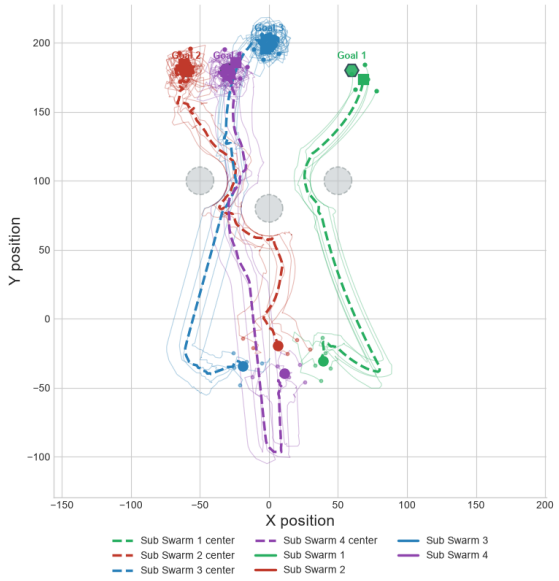


Fig. 4. Driving phase with obstacles: sub-swarm center (dashed) and follower (solid) trajectories, colored by sub-swarm. $M = 8$ leaders, $N = 20$ followers, $K = 4$ goals, 3 obstacles.

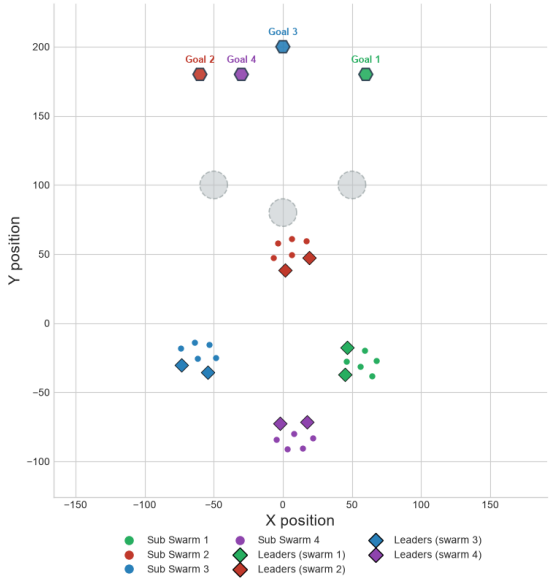


Fig. 5. Driving phase: sub-swarms at different stages relative to their shedding sub-goal. $M = 8$ leaders, $N = 20$ followers, $K = 4$ goals, 3 obstacles.

V. Conclusions and Future Work

We model multi-region SAR deployment as a heterogeneous shepherding problem, where a few capable leaders guide many simple monitoring robots. A single initialization phase assigns tasks evenly. Local follower dynamics and leader behaviors, including collecting, shedding, and driving, then execute these assignments without message exchange. When the common-observation condition is met, deterministic co-shepherd computation assigns unique targets to robots without sharing internal states. The mission ends when compact subgroups reach their assigned regions and begin local coverage and victim detection. Future work will add region selection based on uncertain victim beliefs, and assess robustness to sensing and robot failures.

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